

1. Administrative & Study Identification

Full Study Title: A Phase 2, Multicenter, Double-blind, Two-arm Study of Subcutaneous R07790121 for the Treatment of Subjects With Moderate to Severe Active Crohn's Disease **Short Title/ Acronym:** TAHOE Study **Protocol Number:** GA45392 / 2023-504265-23-00 **NCT Number:** NCT05910528 **Sponsor Name:** Hoffmann-La Roche (formerly Telavant/Roivant) **Investigational Product:** Afimkibart (R07790121, RVT-3101, PF-06480605) **Phase of Development:** Phase 2 **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and efficacy of afimkibart (R07790121) as induction therapy in adult participants with moderately to severely active Crohn's disease (CD), primarily assessed by the proportion of participants achieving clinical remission per Crohn's Disease Activity Index (CDAI) at Week 14. **Secondary Objectives:** To evaluate the proportion of participants achieving an endoscopic response, mucosal healing, and to assess long-term safety and efficacy through the maintenance and open-label extension (OLE) periods.

2.2 Background & Rationale To address the unmet need in patients with moderate to severely active Crohn's disease who do not achieve adequate disease control with current therapies. Afimkibart is a novel monoclonal antibody targeting Tumor Necrosis Factor-like Ligand 1A (TL1A), a cytokine that drives inflammation and fibrosis in immune-mediated disorders. By neutralizing TL1A, the therapy offers a compelling new mechanism of action for patients unresponsive to conventional or advanced therapies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active/Dose-ranging (Two doses of afimkibart) **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 105 participants **Number of Sites:** Global, Multicenter **Estimated Study Start:** July 2023 **Estimated Primary Completion:** 31-Dec-2030

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adults aged 18 to 75 years. 2. Moderately to severely active CD as defined by CDAI and SES-CD, assessed by central read. 3. Elevated very soft or liquid stool frequency and/or abdominal pain. 4. Must have no response, insufficient response, loss of response, and/or intolerance to at least 1 conventional therapy (e.g., corticosteroids) or advanced therapy.

4.2 Exclusion Criteria 1. Diagnosis of ulcerative colitis, indeterminate colitis, microscopic colitis, ischemic colitis, infectious colitis, radiation colitis, or active diverticular disease. 2. Short gut syndrome. 3. Presence of an ostomy or ileoanal pouch. 4. Bowel resection or diversion within ~6 months.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Two different doses of afimkibart administered once a month. **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** 14-week induction period, followed by maintenance and Open-Label Extension (OLE) periods.

5.2 Concomitant & Prohibited Therapy Permitted: Stable doses of conventional CD therapies (e.g., selected aminosalicylates, stable low-dose corticosteroids) as specified by the protocol. **Prohibited:** Concurrent use of other targeted biologics, advanced therapies for Crohn's disease, or live/live-attenuated vaccines during the study.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, baseline CDAI evaluation, Central Read Endoscopy (SES-CD)
Baseline	Day 0	Randomization, First SC Dose of Afimkibart, Vital Signs
Interim	Week 14	Primary Efficacy Assessment (CDAI & Endoscopic Response), Safety Labs
Maintenance/OLE	Post-Week 14	Monthly dosing, long-term safety & efficacy tracking, symptom diaries
End of Study	2030	Final Endoscopy, Exit Interview, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Proportion of participants achieving clinical remission per Crohn's Disease Activity Index (CDAI) at Week 14. **Measurement Method:** Patient-reported outcomes and investigator assessment via the standardized CDAI questionnaire.

7.2 Secondary & Exploratory Endpoints 1. Proportion of participants achieving an endoscopic response (assessed via SES-CD central read) at Week 14. 2. Long-term maintenance of clinical remission and endoscopic healing during the OLE.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection via patient electronic diaries, HCP interviews, and scheduled onsite visits.
Serious Adverse Events (SAEs): Must be reported to the Sponsor and Institutional Review Board within 24 hours of site awareness.
Data Monitoring Committee (DMC): Independent Data Monitoring Committee established to routinely evaluate blinded safety data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all primary and secondary efficacy endpoints; Safety Analysis Set for all AE/SAE evaluations. **Sample Size Justification:** $N=105$ is estimated to provide adequate statistical power to compare the efficacy of the two different dose arms against baseline metrics. **Handling of Missing Data:** Non-Responder Imputation (NRI) utilized for binary clinical remission and endoscopic response endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
Monitoring Plan: Combination of on-site monitoring and centralized remote data monitoring. **Audit Trail:** eCRF data entry with comprehensive audit trails, strict data clarification processes, and central blinded reading for all endoscopic videos.

11. Study Identifiers & Governance

Primary ID: GA45392 **Registry Number:** NCT05910528 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** TAHOE Study **Official Regulatory Title:** A Phase 2, Multicenter, Double-blind, Two-arm Study of Subcutaneous R07790121 for the Treatment of Subjects With Moderate to Severe Active Crohn's Disease

12. Clinical Framework (Crohn's Disease Specific)

Primary Condition: Moderate to Severe Active Crohn's Disease
Diagnosis Threshold: Elevated CDAI score and corresponding SES-CD score via endoscopy **Medical Methodology:** Anti-TL1A targeted

biologic therapy **Optimization Target:** Clinical Remission and Endoscopic Response

13. Study Procedures & Methodology

Management Pathway: Standard induction and maintenance phases for biologic IBD therapy. **Education Protocol (HCP):** Site training on central endoscopy recording requirements and standardized CDAI administration. **Education Protocol (Patient):** Training on reporting stool frequency, abdominal pain scoring, and potential SC injection self-administration if applicable. **Quality Audit Mechanism:** Independent central blinded readers for all endoscopic evaluations (SES-CD) to ensure objective secondary endpoint tracking.

14. Observation & Follow-Up Schedule

Total Duration: Up to 7 years (Estimated completion in 2030 including Open-Label Extension) **Onsite Visit Cadence:** Baseline, Week 14 (Primary Endpoint Assessment), and designated intervals during the OLE. **Remote Monitoring Frequency:** Electronic diary submissions for ongoing CDAI component tracking. **Checklist Review Interval:** Standardized interval reviews for safety labs and concomitant medications.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				